Diabetes Health Indicators

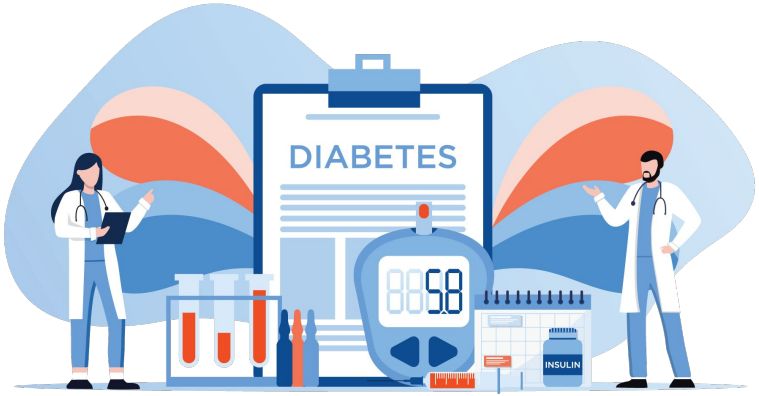
A Statistical Exploration

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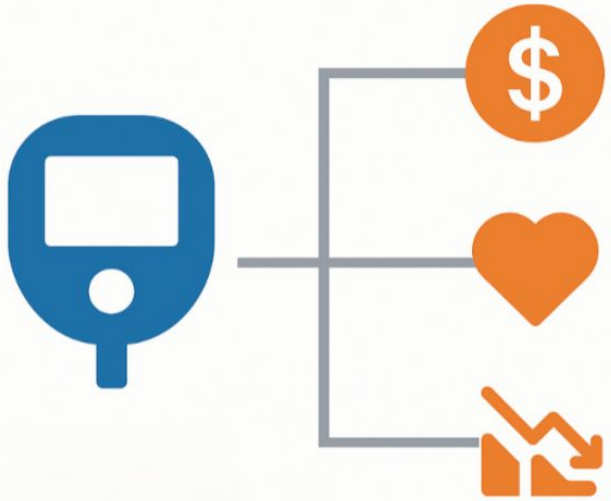
Project Overview and Context

- Diabetes is one of the most common chronic diseases in the U.S.
- Our study focuses on Type II (not genetic)
- Affects millions annually
- Involves impaired blood glucose regulation
- Reduces quality of life and life expectancy
- Is often contracted due to… what exactly?



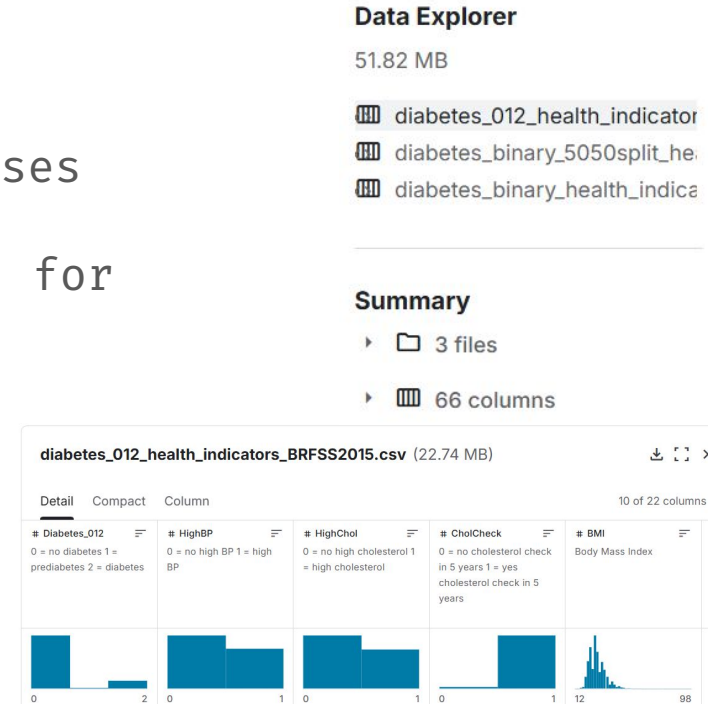
Our Purposes

- To help young people avoid diabetes
- To focus not on determining who *has* diabetes, but who *doesn't*
- To raise awareness of indicators in the control of the individual
- To find patterns between behaviors and demographics
- To avoid pie charts at ALL COSTS



Data Source

- *Diabetes Health Indicators* dataset
  - Kaggle, BRFSS 2015
- Includes over 250,000 survey responses
- Not balanced, which was important for proportional measures
- Covers multiple categories:
  - Health conditions
  - Behaviors
  - Demographic factors

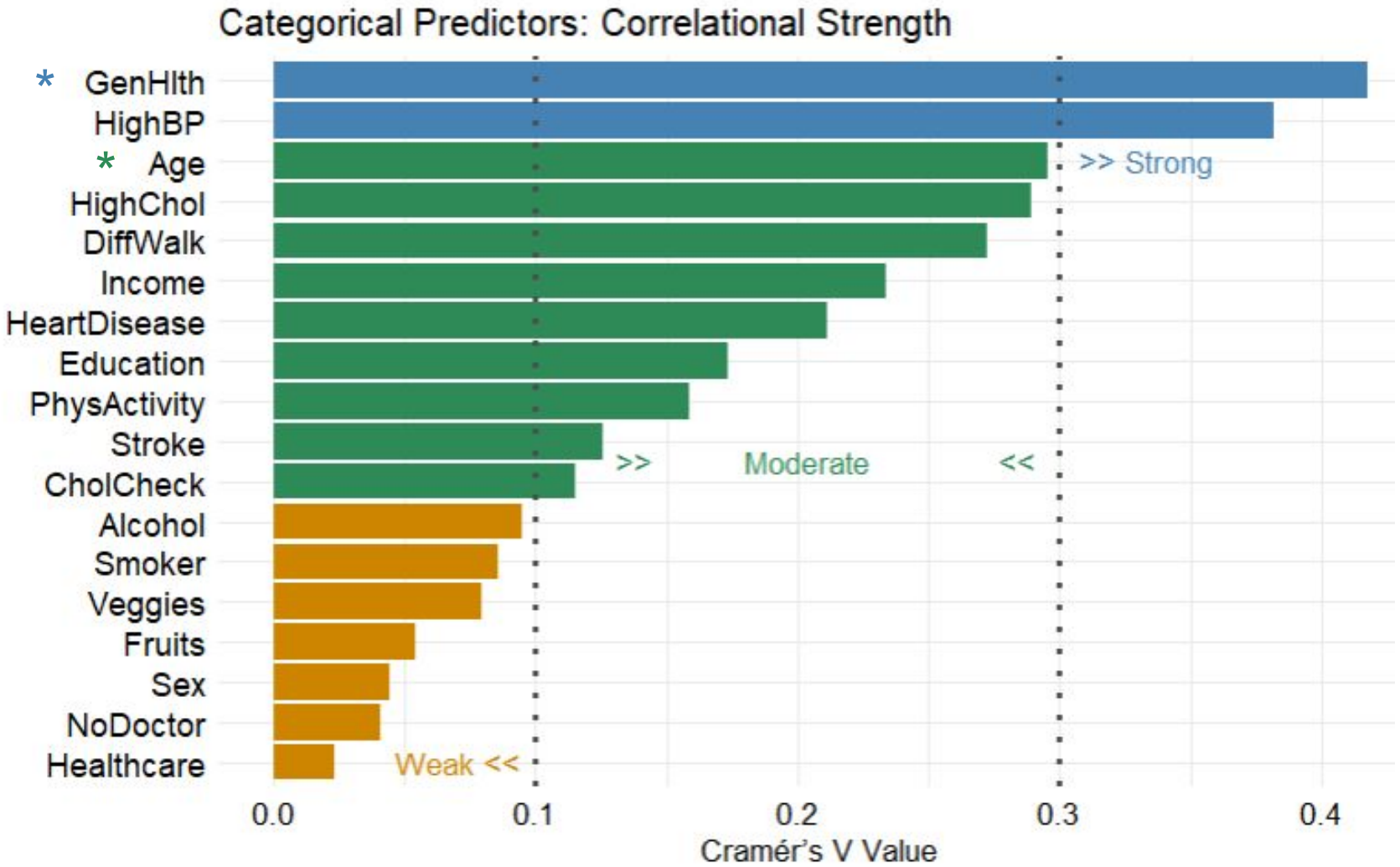


Chi-Square and Cramér's V Analysis

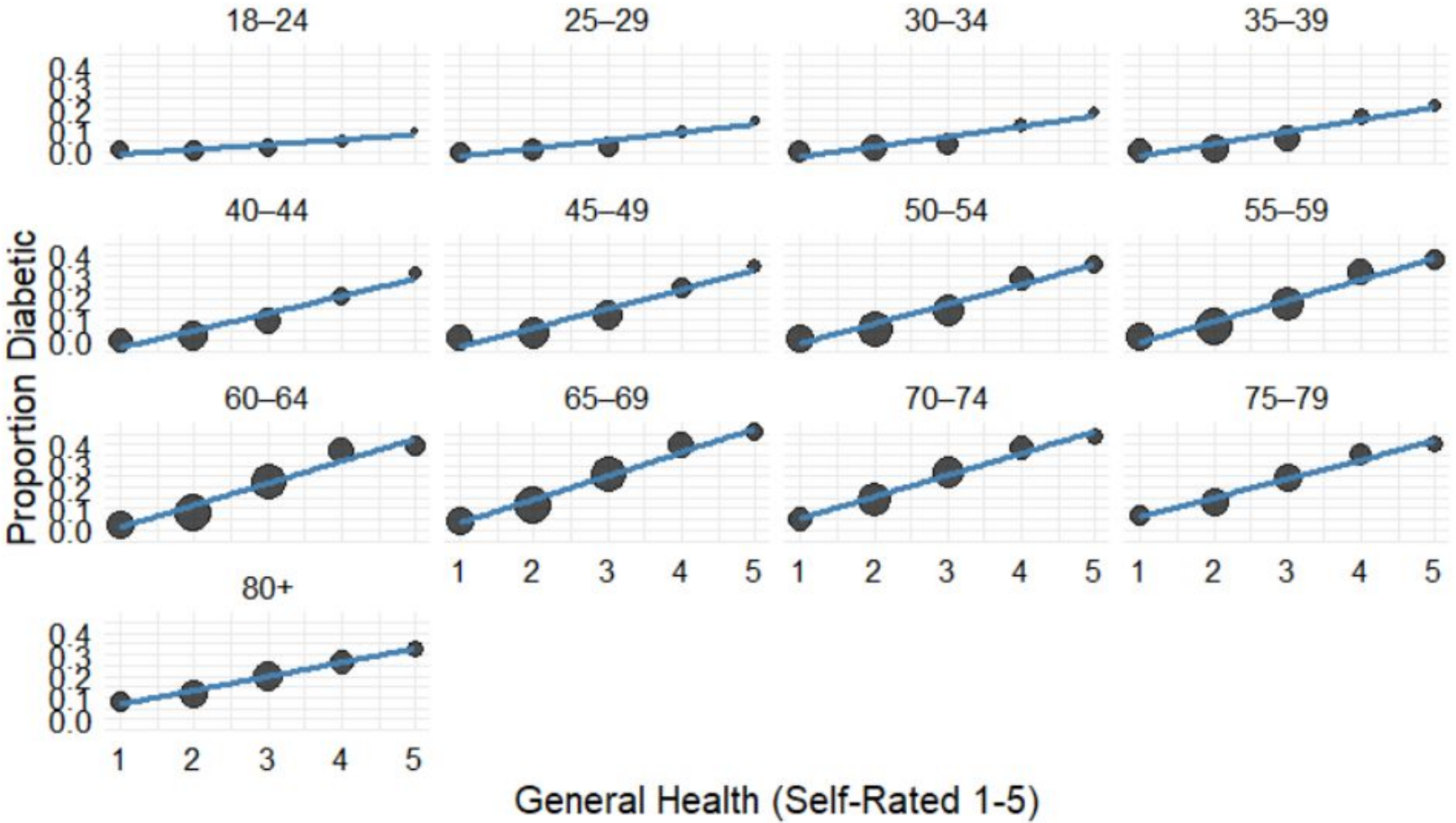
- To determine correlation:
  - Set diabetes as the target
  - Performed Chi-Square analysis in R
  - Standardized values to Cramér's V

- Used a column chart to plot these values

Variable <chr>	Cramers_V <dbl>
HighBP	0.38151555
HighChol	0.28921281
CholCheck	0.11538162
Smoker	0.08599896
Stroke	0.12542678
HeartDiseaseorAttack	0.21152340
PhysActivity	0.15866560
Fruits	0.05407656
Veggies	0.07929315
HvyAlcoholConsump	0.09485314
AnyHealthcare	0.02319075
NoDocbcCost	0.04097657
GenHlth	0.41719938
DiffWalk	0.27264601
Sex	0.04441286
Age	0.29564870
Education	0.17367457
Income	0.23353537


Categorical Predictors: Correlational Strength
* GenHlth
HighBP
* Age
HighChol
DiffWalk
Income
HeartDisease
Education
PhysActivity
Stroke
CholCheck
Alcohol
Smoker
Veggies
Fruits
Sex
NoDoctor
Healthcare
>> Strong
>>
Moderate
<<
Weak <<
0.0
0.1
0.2
0.3
0.4
Cramér's V Value

What Proportion of Each Age Bracket is Diabetic Given Their Health Rating?



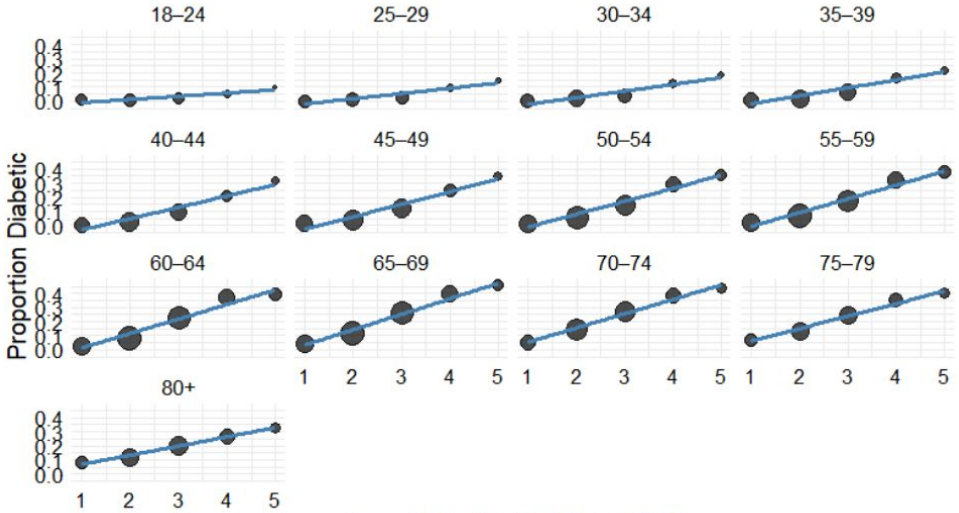
Diabetes Through the Ages

Youth (Our Target Audience):

- Poor general health inferabley due to other factors
- Much lower diabetes rates
- Sampled less

Across all ages:

- Those reporting health scores of 1–3 have ≤25% diabetes rate
- Vast majority with decent health are not diabetic

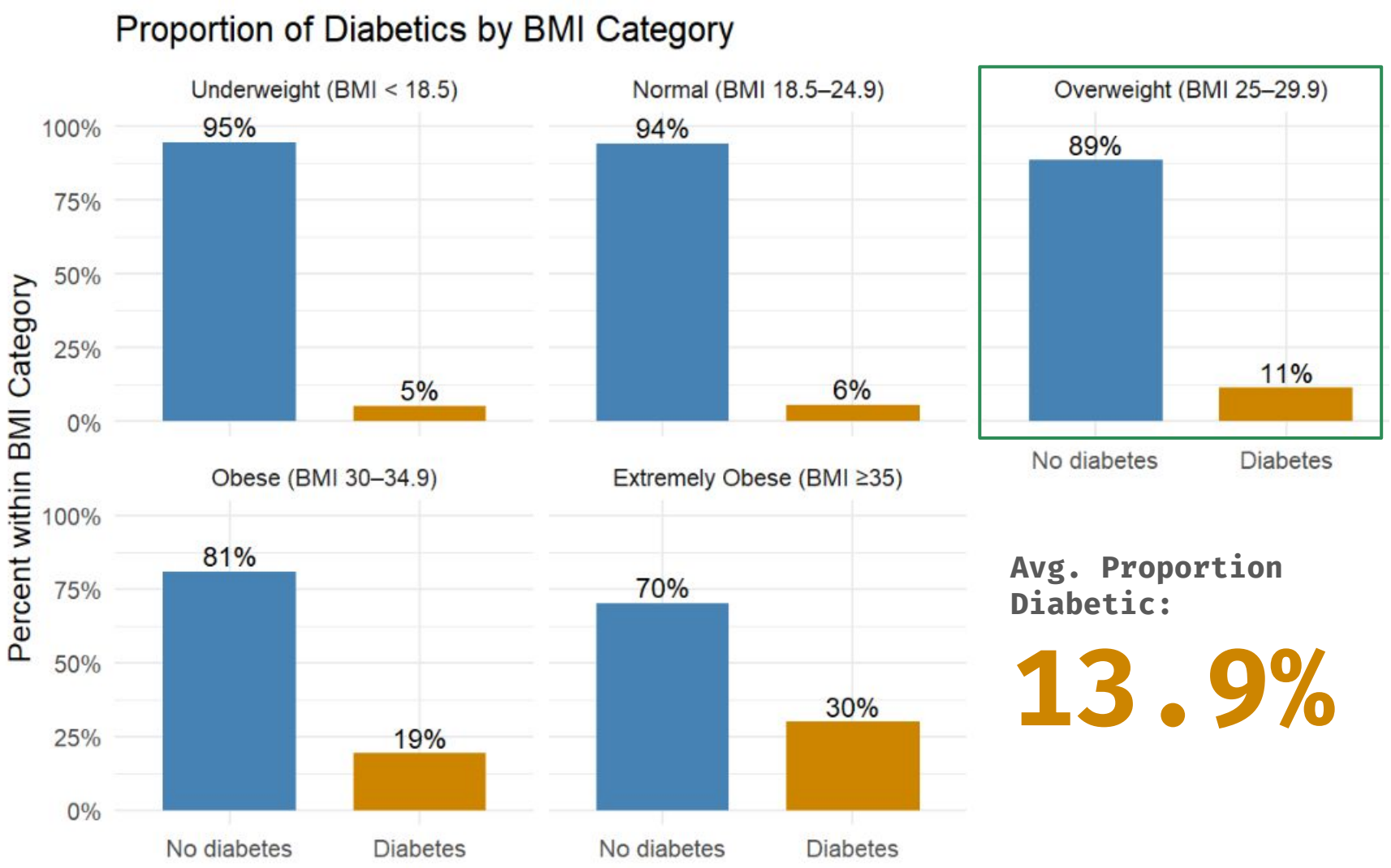


BMI's Relationship to Diabetes

- What about the all-important BMI (Body Mass Index)?
- Diabetics often characterized as overweight
- Conducted simple linear regression
- There is a relationship, but it's not what many of us likely expect…

BMI is a Quantitative-Continuous variable, so it was not included in our Chi-Square Analysis

```
Coefficients:
             Estimate Pr(>|t|)
(Intercept) -0.1831626   <2e-16 ***
BMI          0.0113625   <2e-16 ***
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Signif. codes:  0 '***' 0.001 '**'
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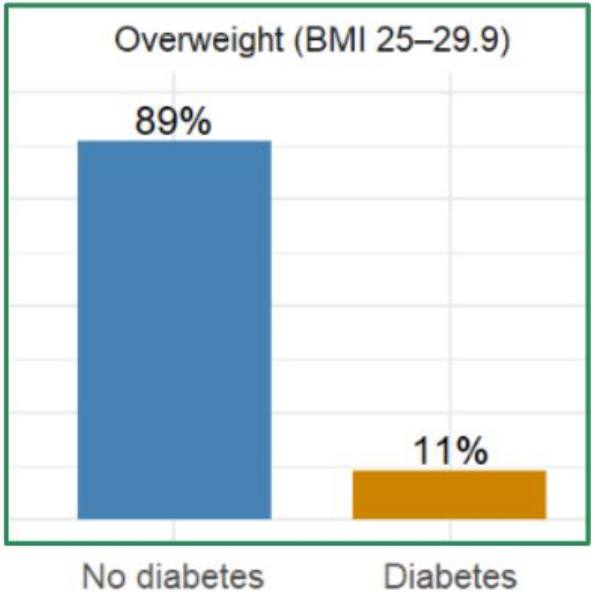


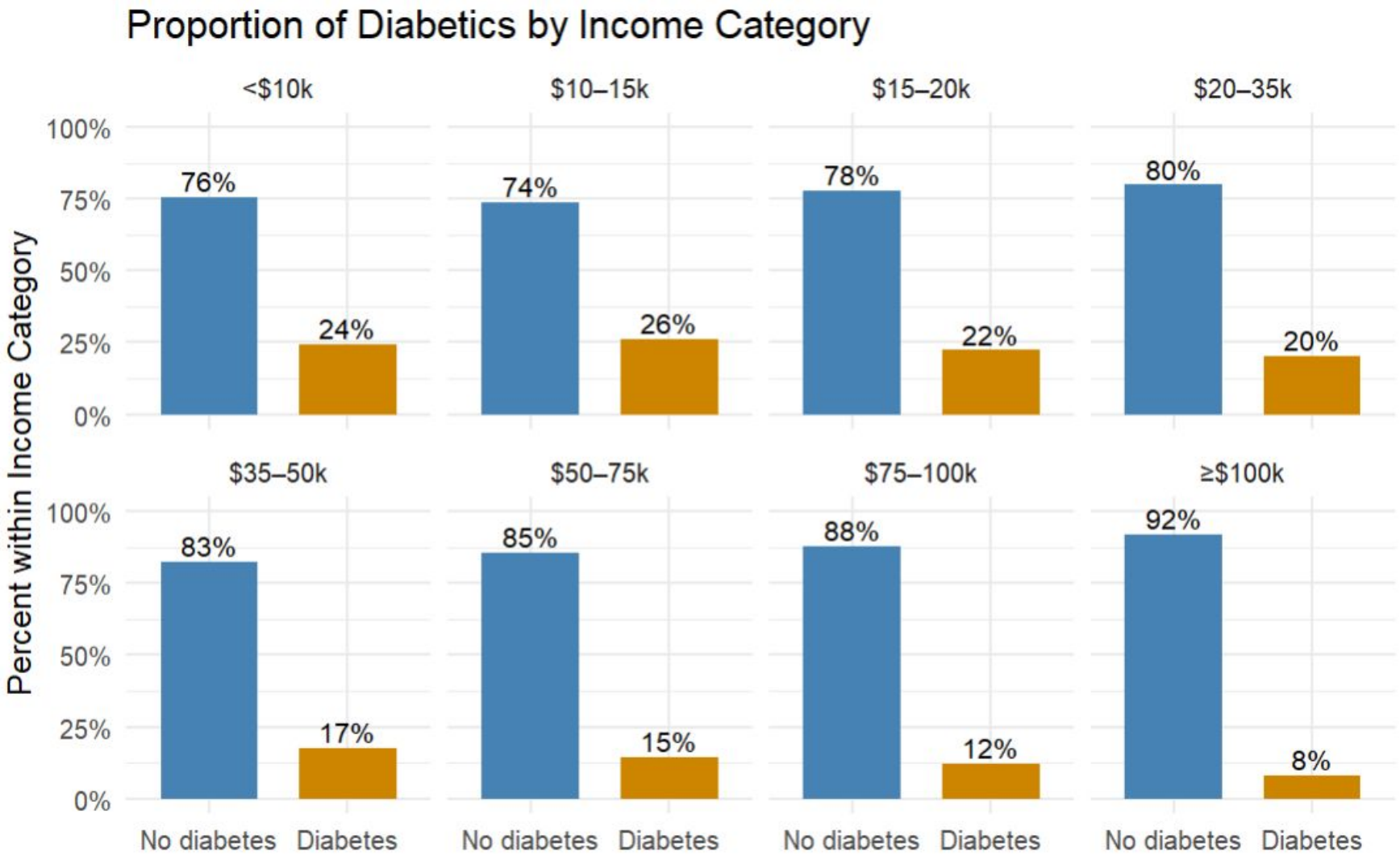
Avg. Proportion Diabetic:

13.9%

What Can We Conclude?

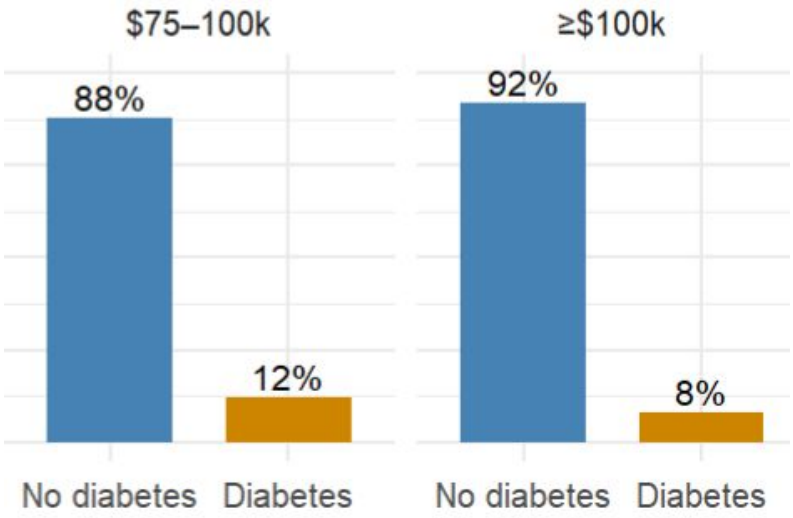
- Is BMI irrelevant then?
- Not at all, BUT it doesn't have the effect many people perceive
- Diabetic rates don't climb above average (~14%) until we reach the "obese" class level
- What other nonbinary categorical indicators (high significance) do we see?




Proportion of Diabetics by Income Category
<$10k
$10–15k
$15–20k
$20–35k
$35–50k
$50–75k
$75–100k
≥$100k
Percent within Income Category
100%
75%
50%
25%
0%
76%
24%
74%
26%
78%
22%
80%
20%
83%
17%
85%
15%
88%
12%
92%
8%
No diabetes
Diabetes

Diabetes Rates Across Income Levels

- More income = lower diabetes rates
- Average of 10% diabetic at the $75,000+ level
- *Even though* they are the oldest age brackets
- Helpful (but expensive) assets:
  - Access to high-quality food
  - Access to health education



Most Significant Predictors

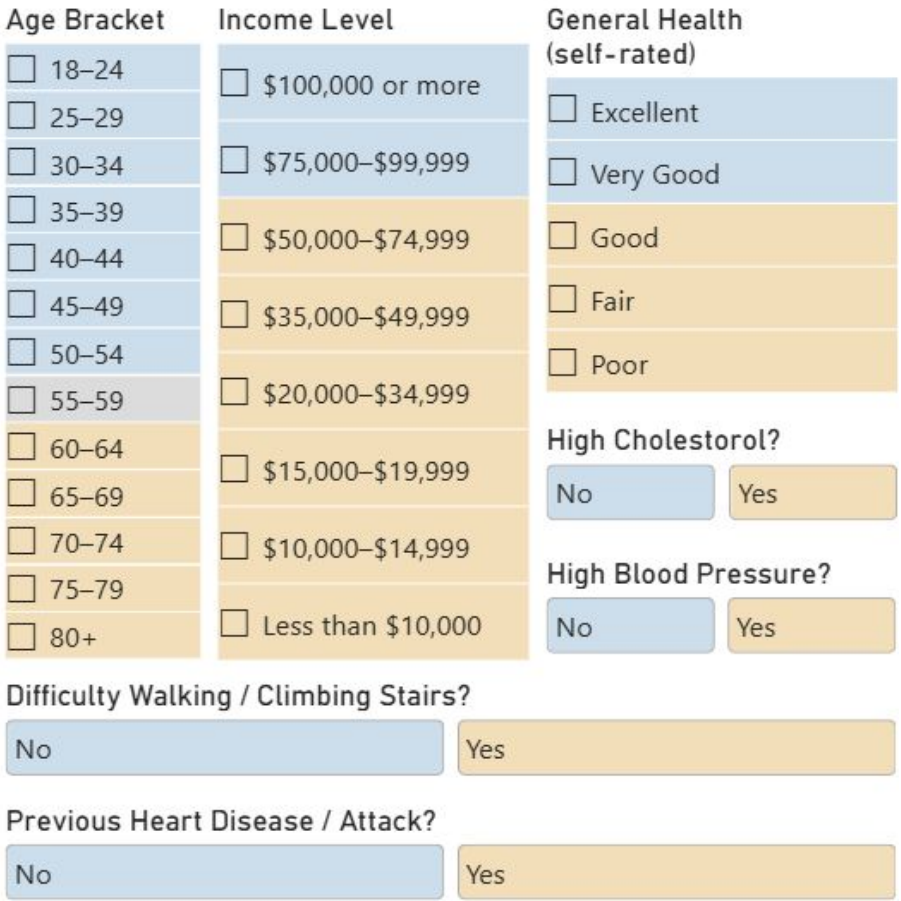
- After filtering our data by the 7 most significant indicators
- Left with the shocking count:

Diabetic Individuals

346 1%

Nondiabetic Individuals

35K 99%



Ideally, we've removed some of the stereotypes surrounding diabetes and replaced them with the activities that can be pursued to actively avoid the disease altogether.

Thank You For Your Time